



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Flow

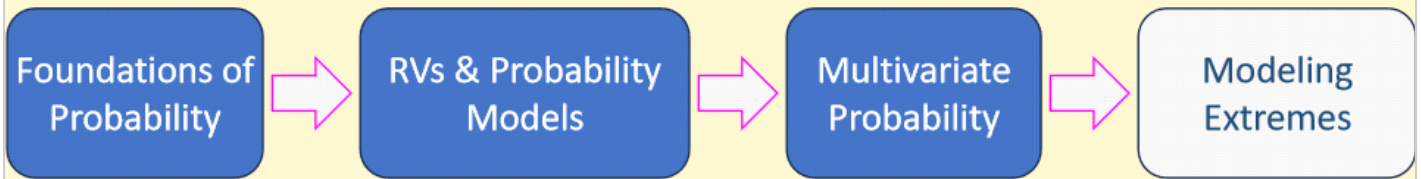
- Announcement
- Recap
- Module-1 starts

Announcement

- Lectures
 - Class tomorrow at 10 am.
- Check GitHub regularly

Course Flow

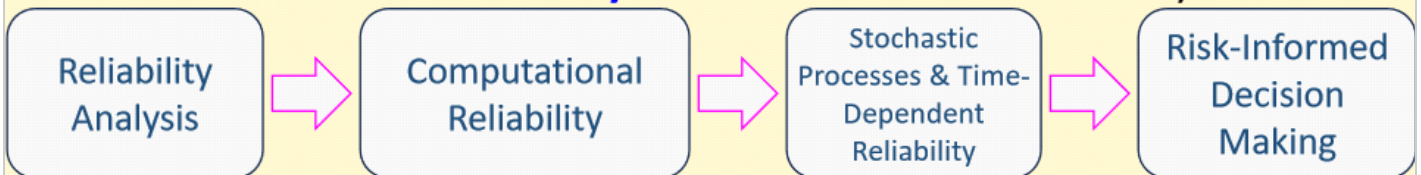
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data

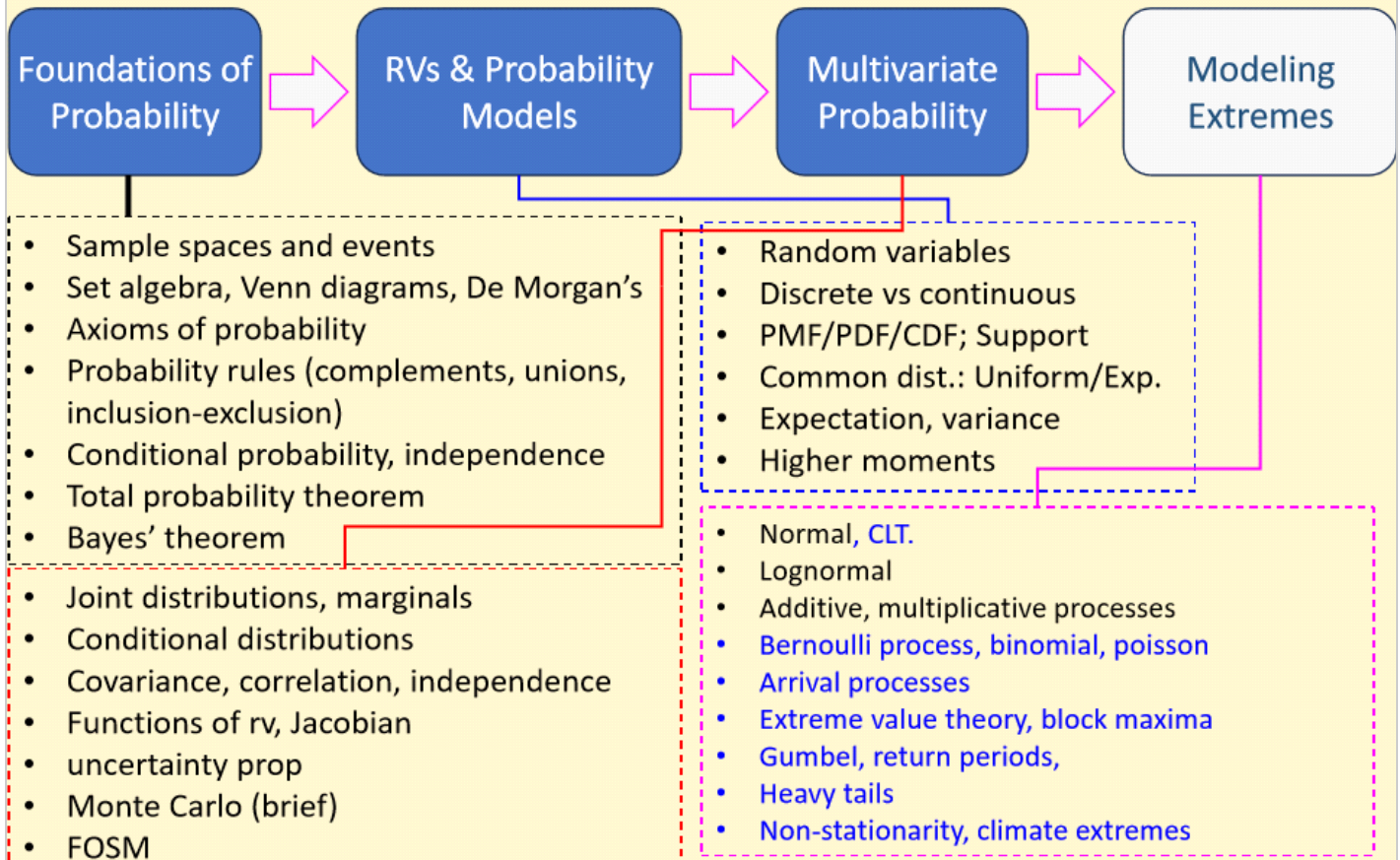


Module 3: **Reliability** – Decision under Uncertainty



Module I: Chapter 1 (M1C1)

Module 1: **Probability** – Understanding Uncertainty



M1C4

Gaussian →

Why Gaussian? (CLT + Lognormal) →

When Gaussian fails (Bernoulli +

Binomial + Poisson) →

Rare-event models →

Extreme-event models (heavy tails)

Motivation

M1C4 lec-1

The Gaussian Distribution: Nature's Default Model

Why is the Gaussian distribution the most important probability model in engineering?

Real-world measurements → Gaussian distribution → Properties → Standardization → Z-score → Normal tables → Engineering applications

Message: The Gaussian is the reference distribution against which much of probability and engineering uncertainty is understood.

M1C4 lec-2

Why the Gaussian? When the Bell Curve Emerges

*Why does the Gaussian appear so frequently in nature—
and when doesn't it?*

Many small additive effects → Central Limit Theorem → Measurement errors → When addition becomes multiplication → Lognormal distribution

Message: Additive randomness leads naturally to the Gaussian; multiplicative randomness often leads to the Lognormal.

L22
Sep 17, 2026

The Central Limit Theorem

- Developed by de Moivre in 1733, but his work was lost for some time
- Gauss independently developed a normal distribution nearly 100 years later

Central Limit Theorem (CLT)

- Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and stdev σ

$$Y = (X_1 + X_2 + \dots + X_n) / n$$

$\mu_Y = \mu$

$n = \text{Number of RVs}$
NOT the number of Rep's

$$\sigma_Y^2 = \left(\frac{1}{n}\right)^2 \sigma_1^2 + \left(\frac{1}{n}\right)^2 \sigma_2^2 + \dots + \left(\frac{1}{n}\right)^2 \sigma_n^2$$

$$\sigma_1 = \sigma_2 = \dots = \sigma_n = \sigma$$

$$\sigma_Y^2 = \sigma^2 / n \Rightarrow \boxed{\sigma_Y = \sigma / \sqrt{n}}$$

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2nd fig.

$$Y_1 = (X_1 + X_2) / 2$$
$$\sigma_{Y_1}^2 = \frac{1}{4} \sigma_1^2 + \frac{1}{4} \sigma_2^2 = \frac{1}{2} \sigma^2$$

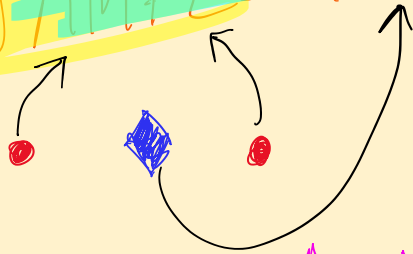
4th fig.

$$Y_2 = (X_1 + X_2 + \dots + X_{10}) / 10$$
$$\sigma_{Y_2}^2 = \frac{1}{10} \sigma^2$$

Central limit theorem

For a "large" sample size, n

SAMPLE MEAN approaches



A NORMAL DISTRIBUTION

With mean μ (same as "population" mean)
AND



stdev $\sigma/\sqrt{n}$ (less than "population" stdev)

M1C4 lec-4

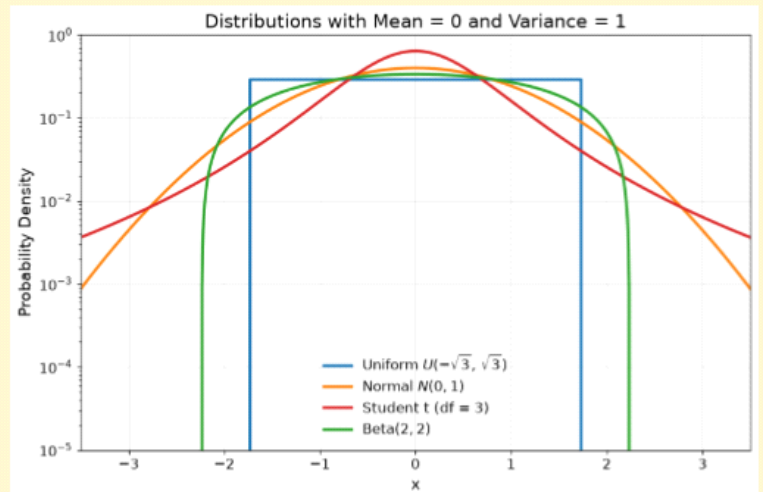
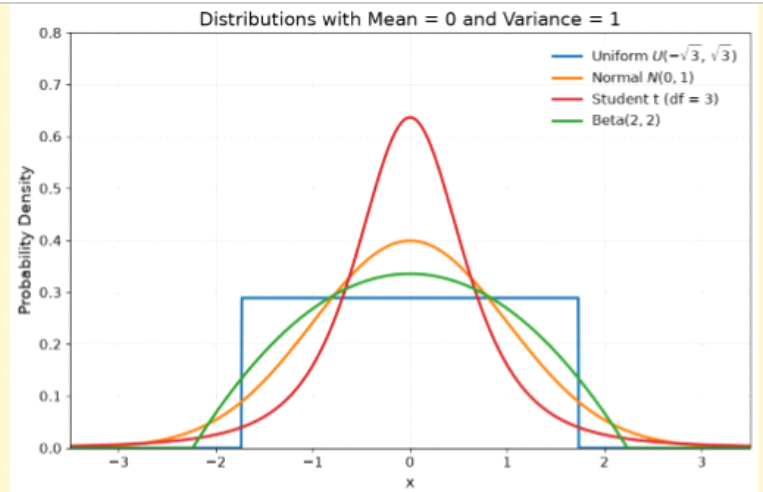
Modeling Extremes

How do we model the largest, most catastrophic events rather than the average ones?

Ordinary distributions fail → Extreme Value Theory →
Return periods → Heavy-tailed distributions → When
Gaussian assumptions become dangerous → Engineering
implications

Message: Engineering failures are often governed not
by typical events, but by the tails of probability
distributions.

Tail of distributions



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Gumbel distributions

- When we are interested in

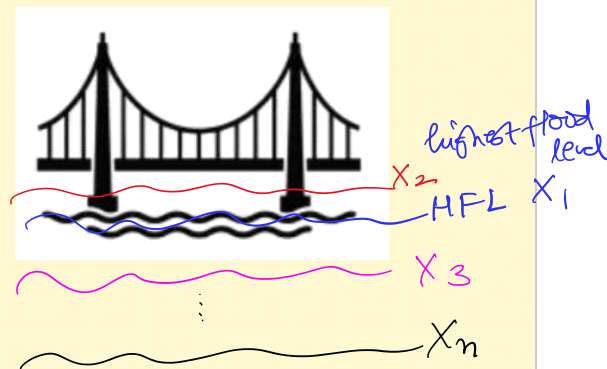
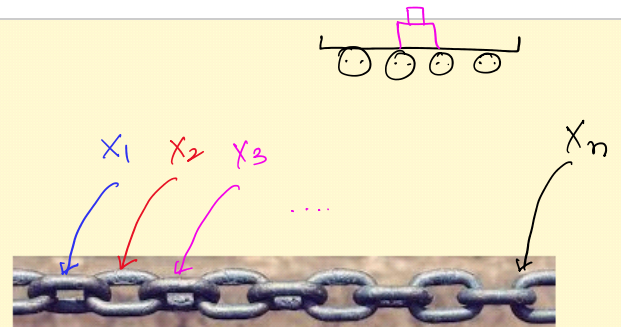
Smallest

OR

Largest

of observation
e.g., chain of links,
flood level under a bridge.

Min. $\min(X_1, X_2, \dots, X_n)$



$$Y_1^{\min} = \min(X_1, X_2, \dots, X_n) \quad \text{~~~~~} X_n$$

$$Y_n^{\max} = \max(X_1, X_2, \dots, X_n)$$

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Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$

$$Y_n = \max(X_1, X_2, \dots, X_n).$$

The CDF of Y_1 is (smallest extreme):

$$F_{Y_1}(y) = \Pr(Y_1 \leq y) = 1 - \Pr(Y_1 > y)$$

$$= 1 - \Pr(X_1 > y \cap X_2 > y \cap \dots \cap X_n > y)$$

$$= 1 - [\Pr(X_1 > y) \cdot \Pr(X_2 > y) \dots \Pr(X_n > y)]$$

CDF

$$= 1 - \prod_{i=1}^n [1 - \Pr(X_i \leq y)]$$

PDF

$$F_{Y_1}(y) = 1 - [1 - F_X(y)]^n$$

$$f_{Y_1}(y) = n [1 - F_X(y)]^{n-1} f_X(y)$$

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Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$
$$Y_n = \max(X_1, X_2, \dots, X_n).$$

The CDF of Y_n is (largest extreme):

$$F_{Y_n}(y) = \Pr(Y_n \leq y)$$
$$= \Pr(X_1 \leq y \cap X_2 \leq y \cap \dots \cap X_n \leq y)$$

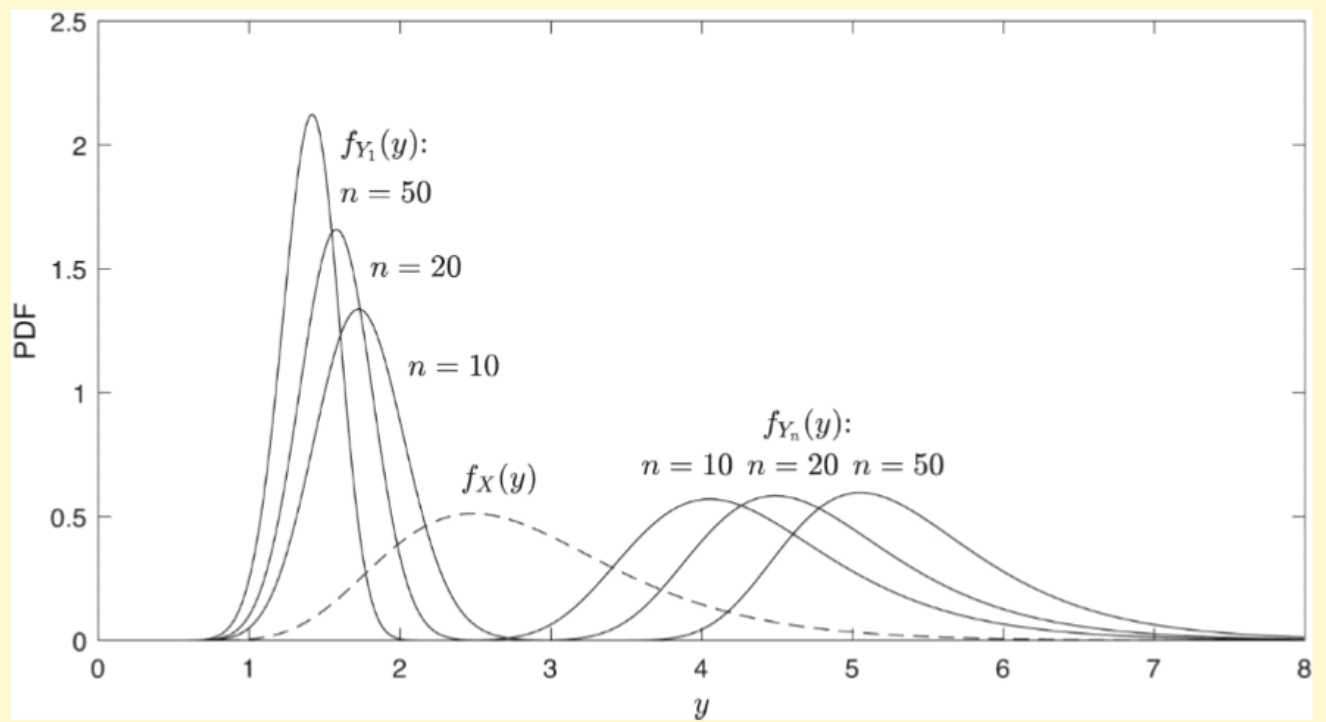
CDF

$$= \prod_{i=1}^n \Pr(X_i \leq y) = [F_X(y)]^n$$

PDF

$$f_{Y_n}(y) = n [F_X(y)]^{n-1} f_X(y)$$

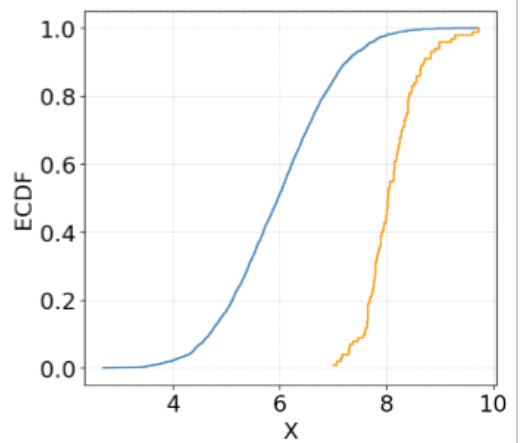
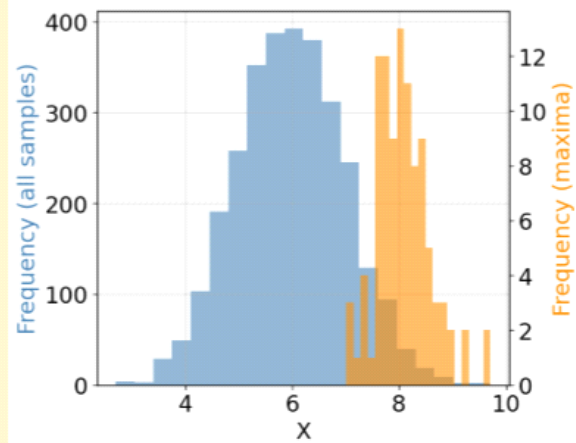
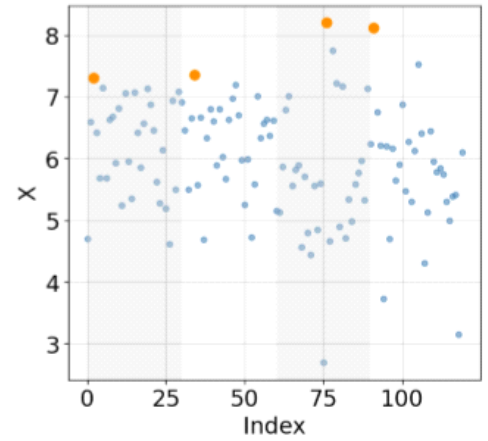
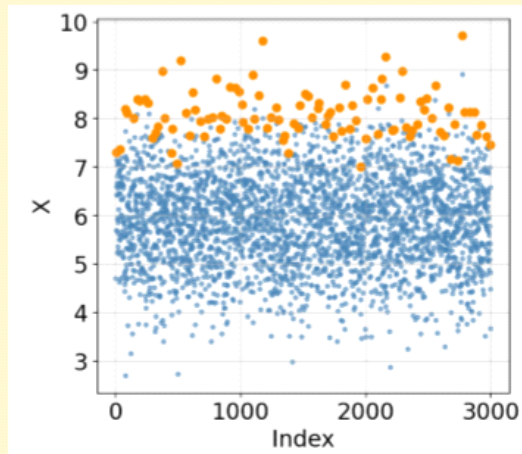
Gumbel distributions



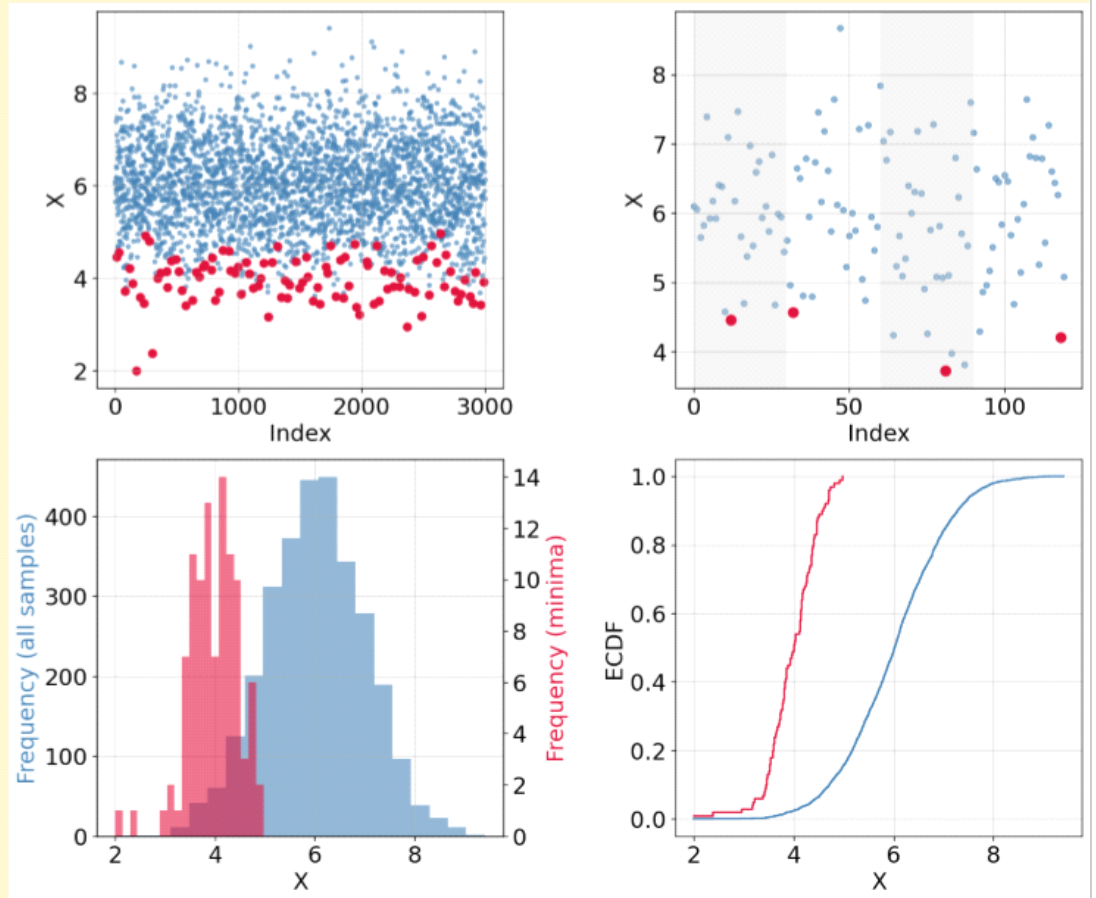
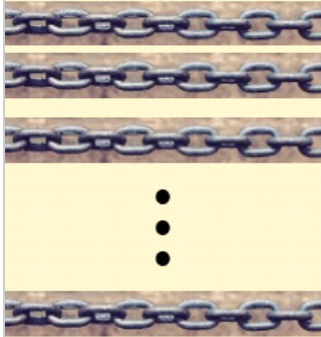
Extreme value analysis



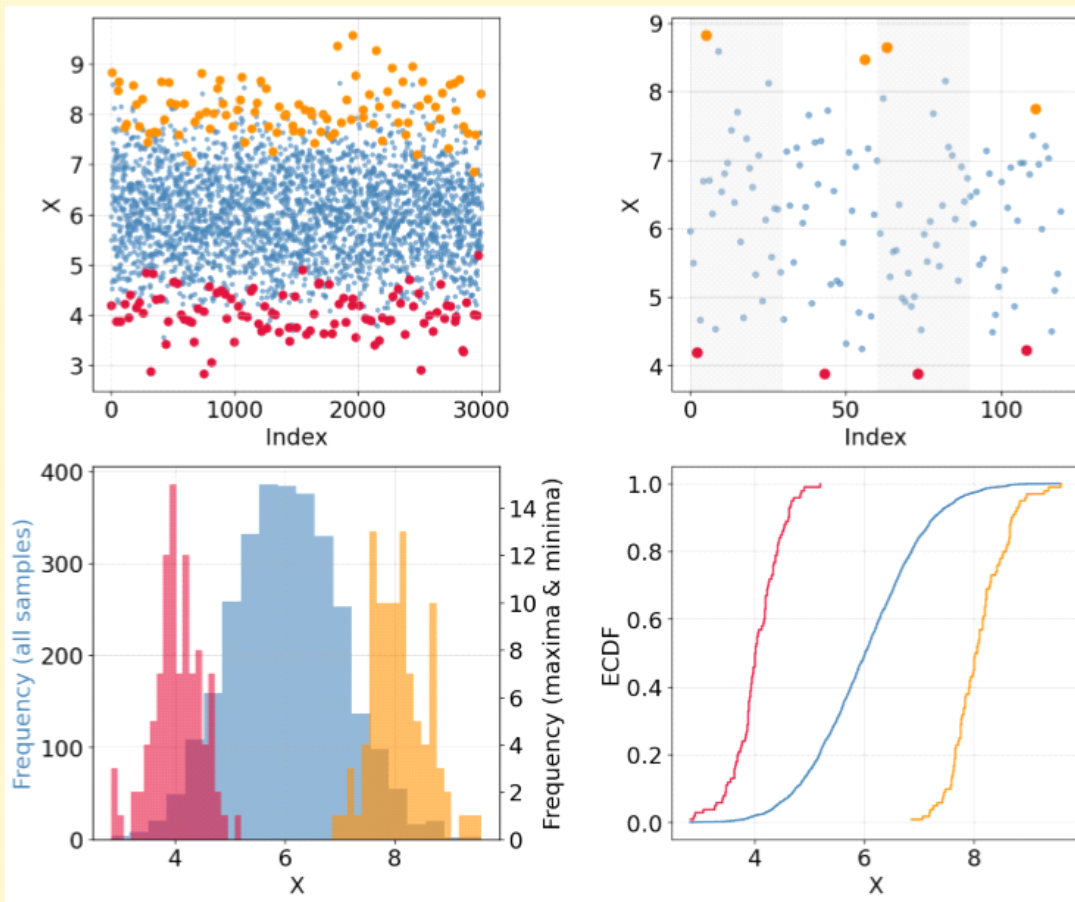
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Extreme value analysis



Extreme value analysis



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Selection of GEV distribution type

Tail type	Extreme value type	Parent distribution

